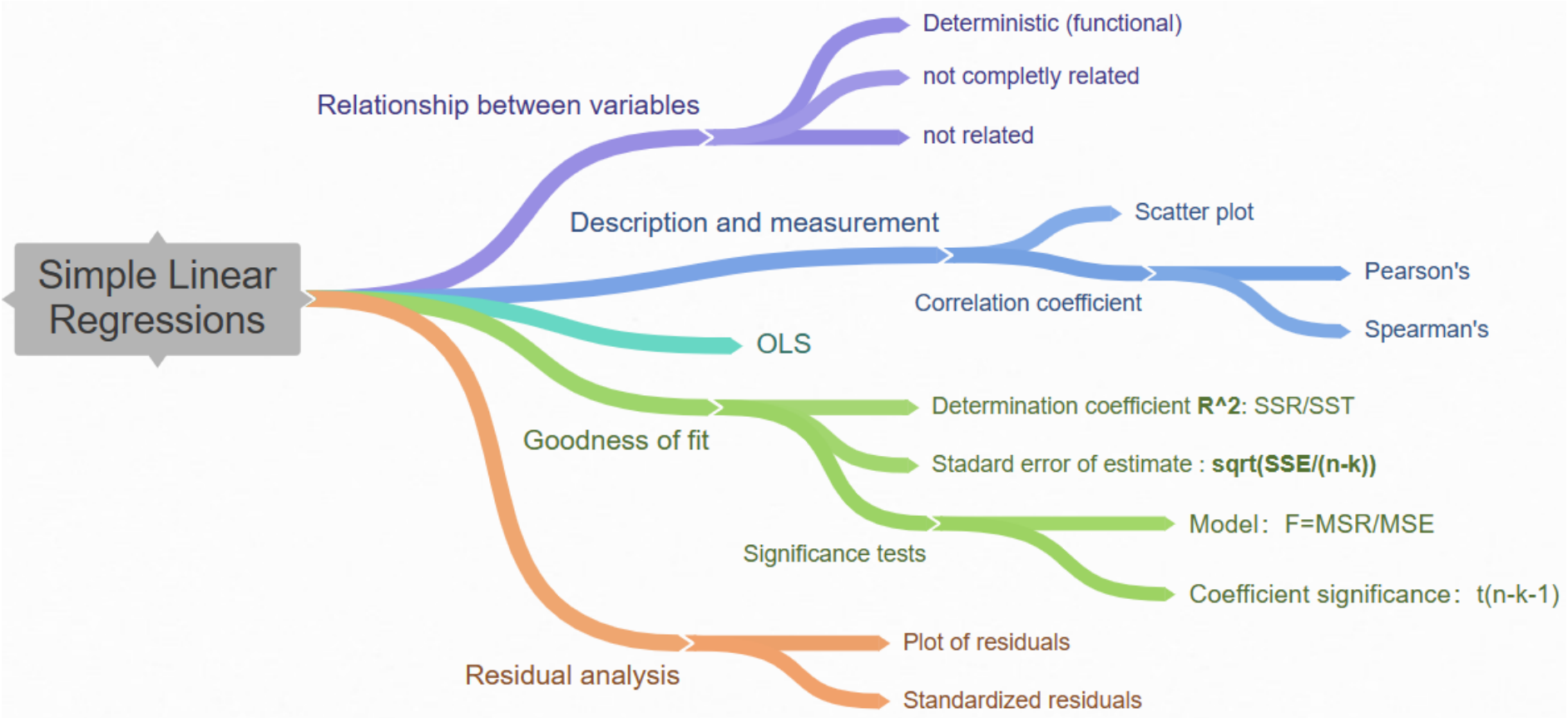
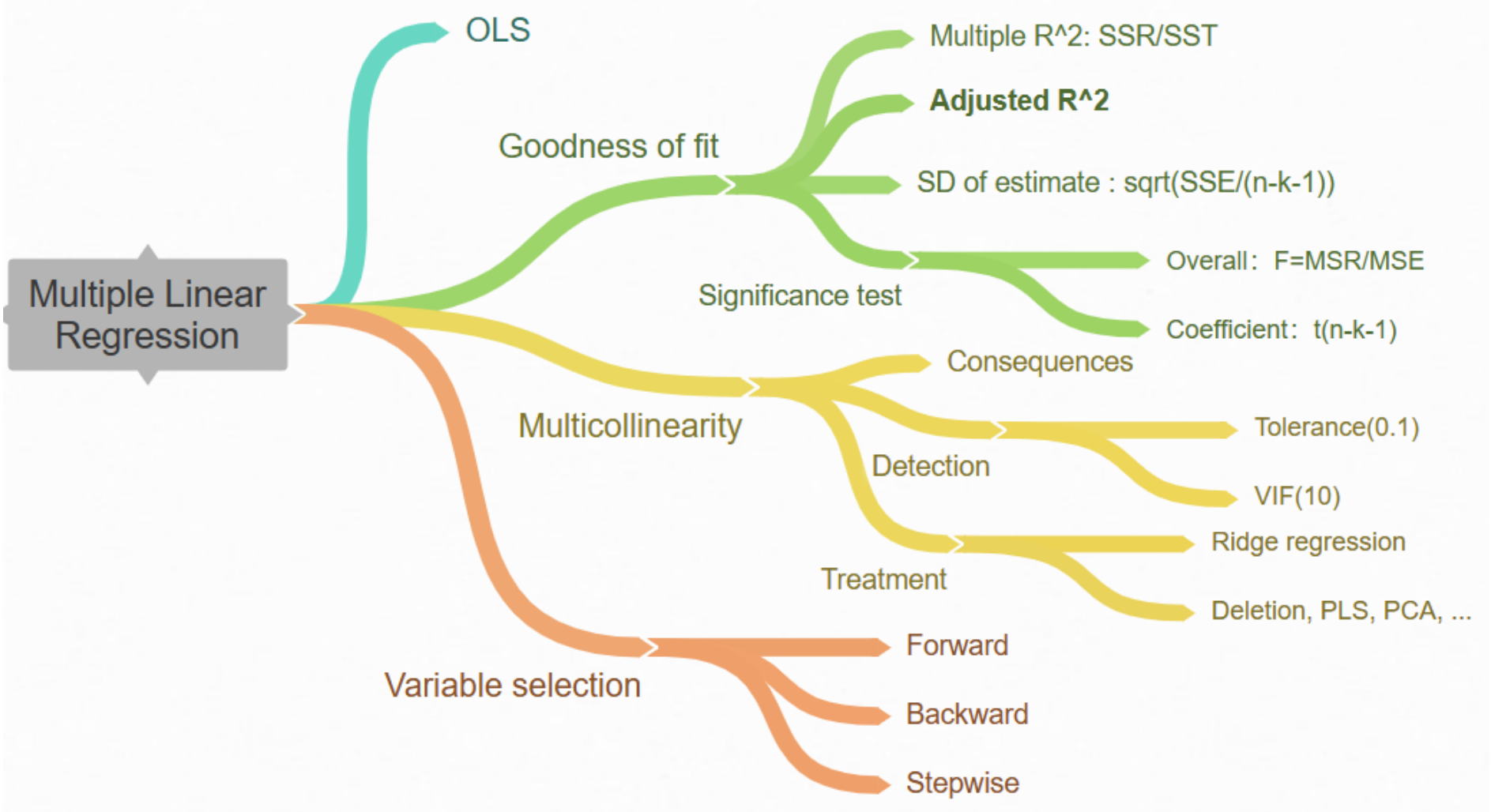
ANalysis Of VAriance (ANOVA)

Jiayun Jin

Review: Regressions



Review: Regressions



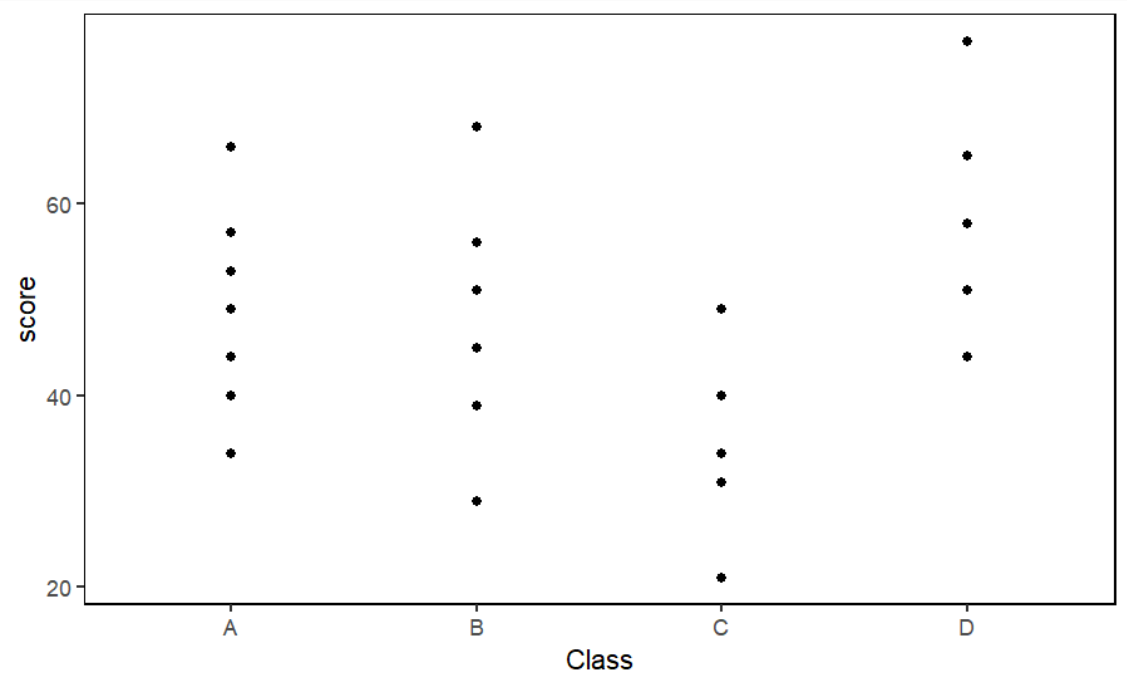
Example: 不同班级的学生成绩

Class A	Class B	Class C	Class D
57	68	31	44
66	39	49	51
49	29	21	65
40	45	34	77
34	56	40	58
53	51		
44			

- 是否可以用班级（分类型变量）来预测学生成绩（数值型变量）？

One-way ANOVA

Scatter plot



- Are the four groups from one population?
- Is there a significant difference among the four groups?

ANOVA

Introduction

- A factor 因子: the categorical variable, e.g., the classes.
- Levels or groups 水平: the different categories within the factor. e.g, the specific classes including A, B, C, and D.
- Number of factors:
  - One-way
  - Two-way
    - Without interactions
    - With interactions

Note: ANOVA can have more than 2 factors. There are benefits, but costs as well as models increase in complexity.

One-way ANOVA

When to use One-way ANOVA

- X: One categorical independent variable
- Y: One quantitative dependent variable

ANOVA tells you if Y changes according to the level of X.

One-way ANOVA

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ANOVA tells you if Y changes according to the level of X.

Examples

We want to find out if there is a difference in

- price per 100 ml among different brand of soda (Coke, Pepsi, Sprite, and Fanta)
- hours of sleep among groups of people having different levels of social media use (low, medium, high)

One-way ANOVA

The hypothesis

- H_0: There is no difference among group means
- H_1: At least one group differs significantly from the overall mean of the dependent variable

One-way ANOVA

Assumptions

1. Independence of observations

   Observations in each group are independent from each other

2. Normality

   In each group, observations are drawn from a normally distributed population

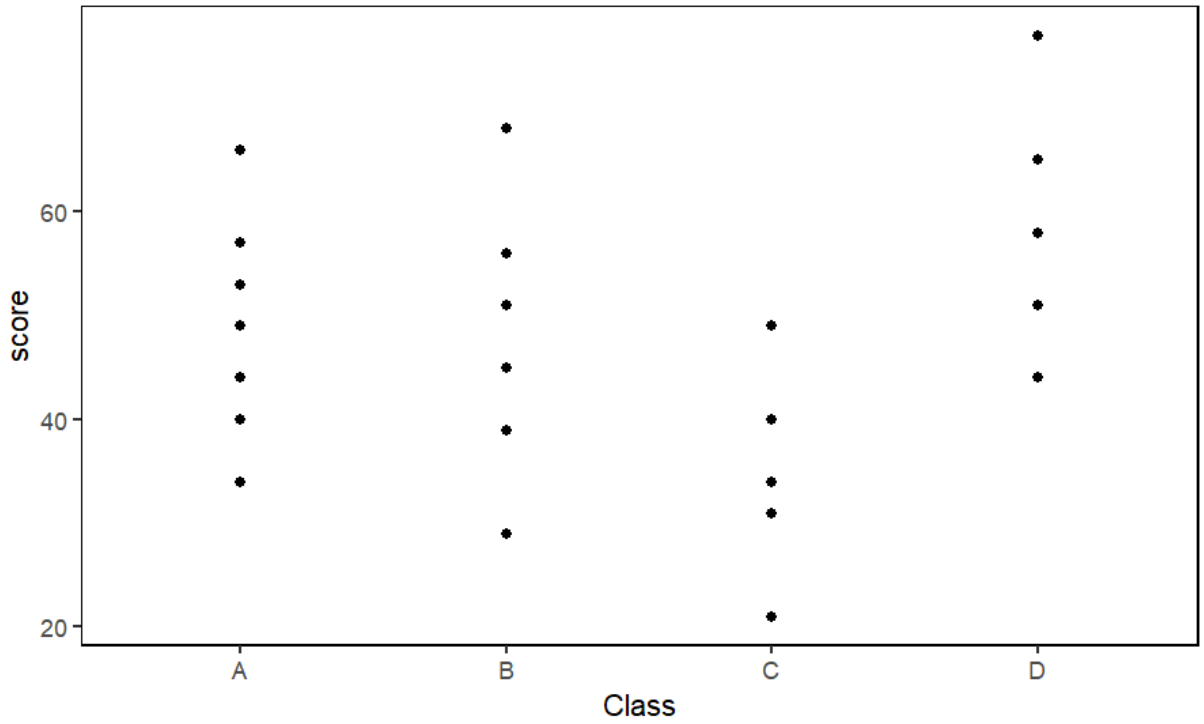
3. Equal variances

   The variances of the population that the samples come from in each group are equal

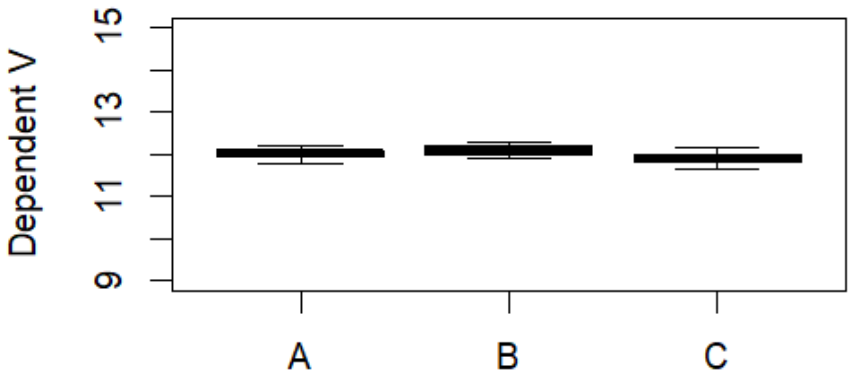
Note: Based on the above assumptions, the task is to examine whether the means of four Normally-distributed populations with identified variance are different.

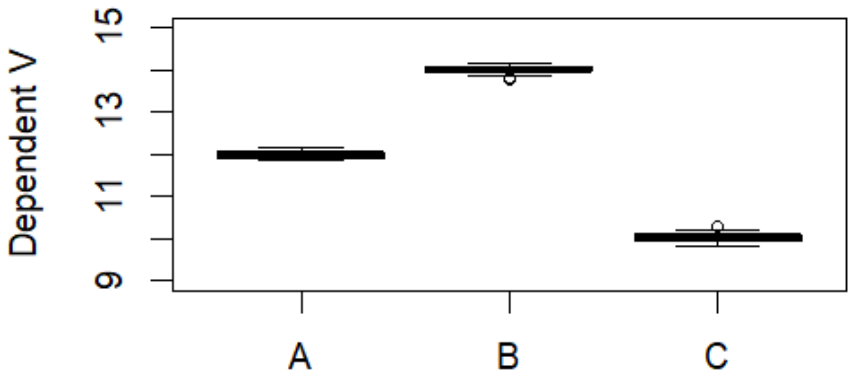
One-way ANOVA: How does it work

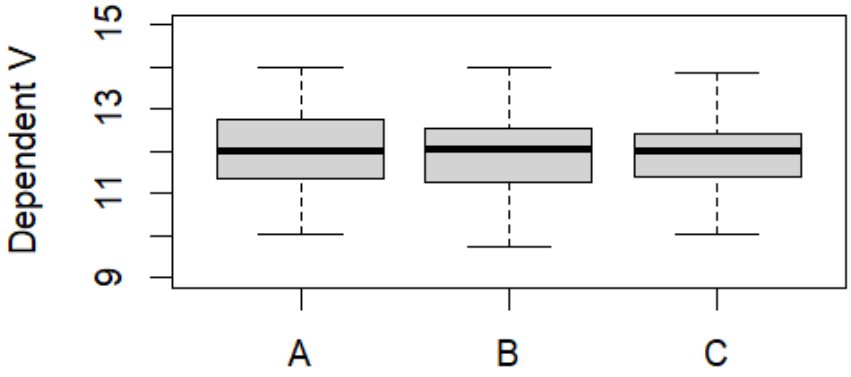
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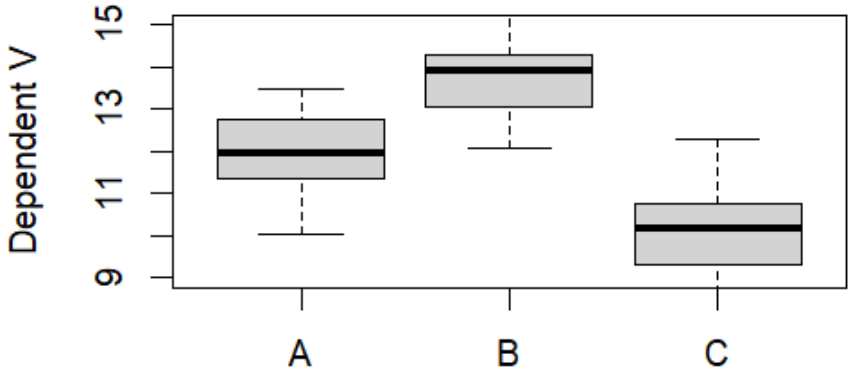


Within-group variance and between-group variance

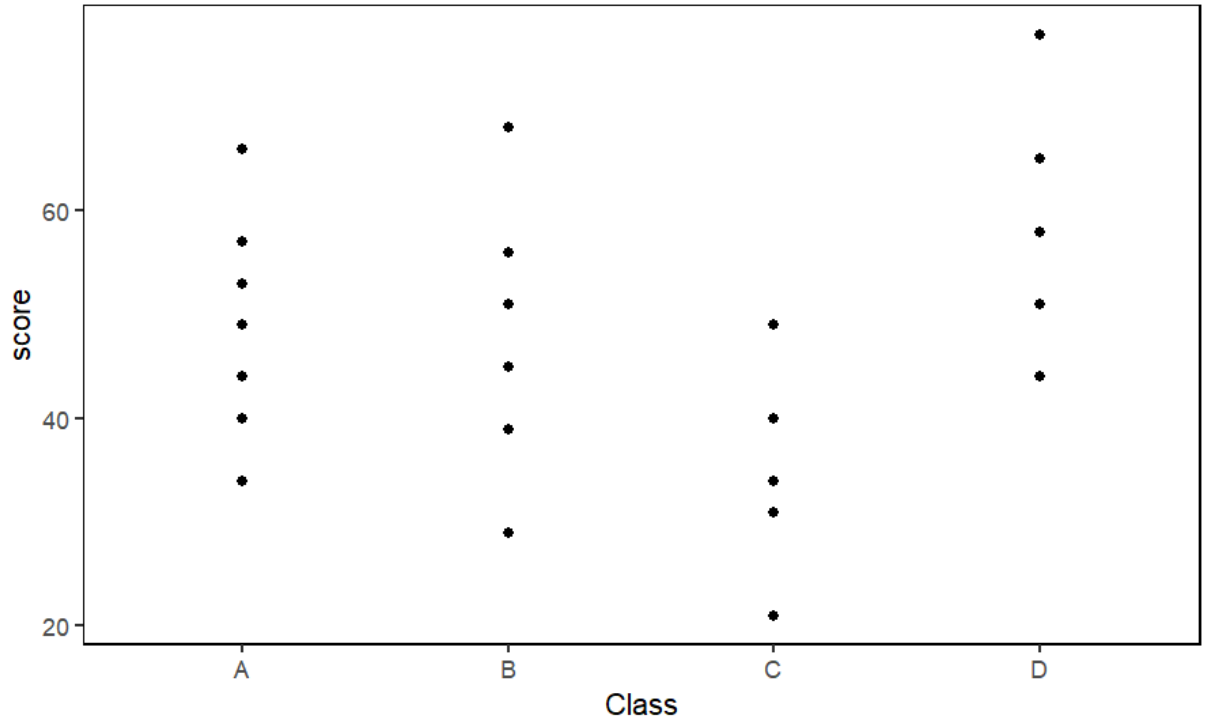








A	B	C	D
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Decomposition of variation

1. Variation within groups: SSE
2. Variation between groups: SSR
3. Total variation: SST = SSE + SSR where SS stands for sum of squares
4. 如果不同班级之间的学生成绩没有显著差异, SSR/SSE 应该比较小.

The procedure (I)

1. The hypothesis

$H_0 : \mu_1 = \mu_2 = ... = \mu_k$

H_1 : at least one population mean is different from the others

2. F statistic
   - Calculate the group means $\bar{x_1}, \bar{x_2}, ..., \bar{x_k}$

$$\bar{x_i} = \frac{\sum_{j=1}^{n_i} x_{ij}}{n_i}, i = 1, 2, ..., k$$

   - Estimate the overall mean $\bar{\bar{x}}$

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{k} \sum_{j=1}^{n_i} x_{ij}}{n} = \frac{\sum_{i=1}^{k} n_i \bar{x_i}}{n}$$

The procedure (II)

- Calculate sum of squares
  1. Total sum of squares

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{\bar{x}})^2$$

  2. Sum of squares between groups

$$SSR = \sum_{i=1}^{k} n_i (\bar{x}_i - \bar{\bar{x}})^2$$

  3. Sum of squares withing groups

$$SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

The procedure (III almost there...)

- F statistic
  1. Mean square of variance between groups

$$MSR = \frac{\text{Sum of squares between groups}}{df} = \frac{SSR}{k-1}$$

  2. Mean square of variance within groups

$$MSE = \frac{\text{Sum of squares within groups}}{df} = \frac{SSE}{n-k}$$

  3. Test statistic

$$F = \frac{MSR}{MSE} \sim F(k-1, n-k)$$

3. Decision
   - if $F > F_\alpha(k-1, n-k)$, $H_0(\mu_1 = \mu_2 = ... = \mu_k)$ is rejected
   - if $F < F_\alpha(k-1, n-k)$, $H_0 : \mu_1 = \mu_2 = ... = \mu_k$ is not rejected

Back to the example

A	B	C	D
57	68	31	44
66	39	49	51
49	29	21	65
40	45	34	77
34	56	40	58
53	51		
44			
$\bar{x}_1 = 49$	$\bar{x}_2 = 48$	$\bar{x}_3 = 35$	$\bar{x}_4 = 59$
$n_1 = 7$	$n_2 = 6$	$n_3 = 5$	$n_4 = 5$

1. $\bar{\bar{x}} = 47.87$
2. $SSR = 7\times(49-47.87)^2+6\times(48-47.87)^2+5\times(35-47.87)^2+5\times(59-47.87)^2$
3. $SSE = 700 + 924 + 434 + 650$
4. $F = \frac{SSR/(k-1)}{SSE/(n-k)} = 3.406$

One-way ANOVA

One-way ANOVA table

Source	Sum of squares (SS)	df	Mean squares (MS)	F
Between groups	SSR	k-1	MSR	MSR/MSE
Within groups	SSE	n-k	MSE	
Total	SST	n-1		

Strength of the relationship

$$R^2 = \frac{SSR}{SST}$$

Multiple comparisons

The motivation

- ANOVA only tells us ***whether any group means differ from another***, but does not tell us ***which groups differ***.

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Tukey's Honest Significant Differences (HSD)

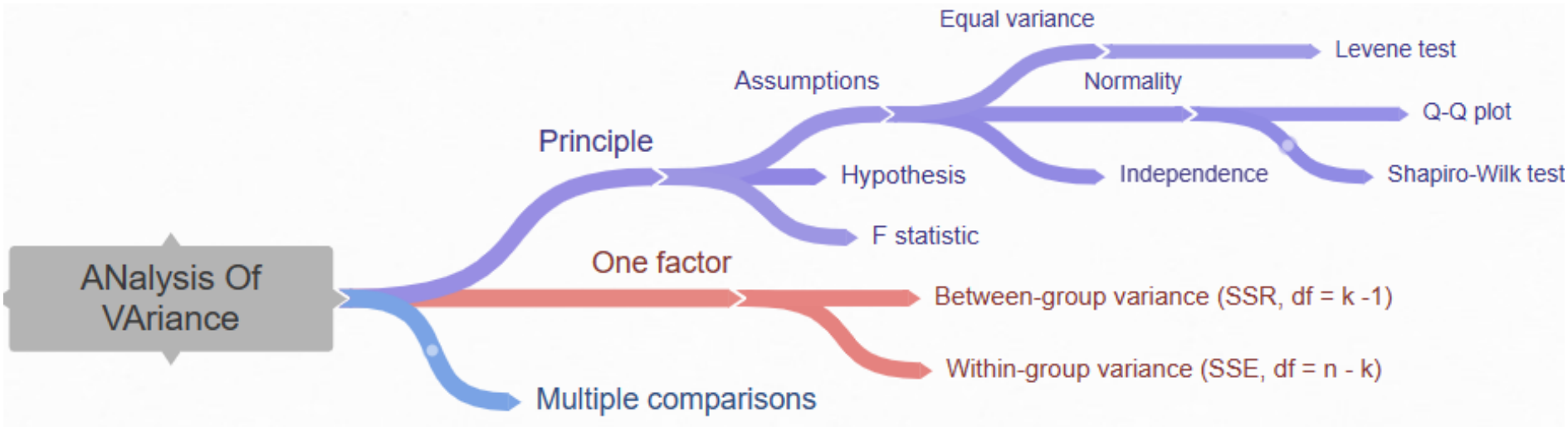
$$H_0 : u_i = u_j$$
$$H_1 : u_i \neq u_j$$

where i and j represent two different populations.

$$|\bar{x}_i - \bar{x}_j|$$

with an overall significance level of α (all pairwise tests can be run at the same time with an overall significance level of α)

Summary


ANalysis Of VAriance
Principle
Assumptions
Equal variance
Levene test
Normality
Q-Q plot
Shapiro-Wilk test
Independence
Hypothesis
F statistic
One factor
Between-group variance (SSR, df = k -1)
Within-group variance (SSE, df = n - k)
Multiple comparisons

Exercise

我们需要分析不同种植条件下植物的生长情况。

在三种不同的光照条件下，全光照、半阴、全阴，我们记录了植物的生长高度。

试确定：不同光照条件下植物生长高度是否存在显著差异？

1. 列出原假设
2. 生成数据 (基于 Markdown 文件)
3. 执行 anova
4. 解读结果（F 和 p）